Post-Route STA - vsdbabysoc (16 PVT Corners)

Library: sky130_fd_sc_hd | Tool: OpenROAD/OpenSTA | Parasitics: estimated from routed design

Corner	WNS	TNS	Worst Setup	Worst Hold
ff_100C_1v65	0.00	0.00	7.44	0.26
ff_100C_1v95	0.00	0.00	8.26	0.21
ff_n40C_1v56	0.00	0.00	6.35	0.31
ff_n40C_1v65	0.00	0.00	7.01	0.27
ff_n40C_1v76	0.00	0.00	7.57	0.24
ff_n40C_1v95	0.00	0.00	8.21	0.20
ss_100C_1v40	-2.64	-340.90	-2.64	0.91
ss_100C_1v60	0.00	0.00	1.76	0.65
ss_n40C_1v28	-27.02	-12774.20	-27.02	1.67
ss_n40C_1v35	-14.70	-5623.47	-14.70	1.26
ss_n40C_1v40	-9.52	-2787.18	-9.52	1.06
ss_n40C_1v44	-6.50	-1500.01	-6.50	0.95
ss_n40C_1v60	0.00	0.00	0.20	0.64
ss_n40C_1v76	0.00	0.00	3.28	0.49
tt_025C_1v80	0.00	0.00	6.42	0.32
tt_100C_1v80	0.00	0.00	6.48	0.32

Corner: ff_100C_1v65

```
Startpoint: core.CPU_src1_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_src2_value_a3[20]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.66    0.66   clock network delay (propagated)
   0.00    0.66 ^ core.CPU_src1_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.33    0.99 ^ core.CPU_src1_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.08    1.07 v _05408_/Y (sky130_fd_sc_hd__inv_1)
   0.21    1.28 v _10738_/COUT (sky130_fd_sc_hd__ha_1)
   0.36    1.64 v _10735_/COUT (sky130_fd_sc_hd__fa_1)
   0.14    1.78 ^ _07426_/Y (sky130_fd_sc_hd__nor2b_1)
   0.10    1.88 v _07428_/Y (sky130_fd_sc_hd__o21bai_1)
   0.27    2.15 ^ _07431_/Y (sky130_fd_sc_hd__a31oi_1)
   0.24    2.39 v _07553_/Y (sky130_fd_sc_hd__o2111ai_1)
   0.50    2.89 ^ _07920_/Y (sky130_fd_sc_hd__a21boi_0)
   0.11    3.00 v _07923_/Y (sky130_fd_sc_hd__nor3_1)
   0.24    3.24 ^ _07924_/Y (sky130_fd_sc_hd__xnor2_1)
   0.13    3.38 v _07934_/Y (sky130_fd_sc_hd__a221oi_1)
   0.20    3.58 v place248/X (sky130_fd_sc_hd__buf_4)
   0.14    3.72 v _09825_/X (sky130_fd_sc_hd__and2_1)
   0.41    4.13 ^ _10170_/Y (sky130_fd_sc_hd__mux2i_1)
   0.00    4.13 ^ core.CPU_src2_value_a3[20]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           4.13   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.70   11.70   clock network delay (propagated)
   0.00   11.70   clock reconvergence pessimism
          11.70 ^ core.CPU_src2_value_a3[20]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.12   11.57   library setup time
          11.57   data required time
---------------------------------------------------------
          11.57   data required time
          -4.13   data arrival time
---------------------------------------------------------
           7.44   slack (MET)


Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.66    0.66   clock network delay (propagated)
   0.00    0.66 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.24    0.90 ^ core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.90 ^ core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_4)
           0.90   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.66    0.66   clock network delay (propagated)
   0.00    0.66   clock reconvergence pessimism
           0.66 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_4)
  -0.02    0.64   library hold time
           0.64   data required time
---------------------------------------------------------
           0.64   data required time
          -0.90   data arrival time
---------------------------------------------------------
           0.26   slack (MET)


wns max 0.00
tns max 0.00
worst slack max 7.44
```

```
worst slack min 0.26
```

Corner: ff_100C_1v95

```
Startpoint: core.CPU_src1_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_src2_value_a3[20]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.54    0.54   clock network delay (propagated)
   0.00    0.54 ^ core.CPU_src1_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.26    0.80 ^ core.CPU_src1_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.06    0.87 v _05408_/Y (sky130_fd_sc_hd__inv_1)
   0.17    1.04 v _10738_/COUT (sky130_fd_sc_hd__ha_1)
   0.28    1.31 v _10735_/COUT (sky130_fd_sc_hd__fa_1)
   0.11    1.43 ^ _07426_/Y (sky130_fd_sc_hd__nor2b_1)
   0.08    1.51 v _07428_/Y (sky130_fd_sc_hd__o21bai_1)
   0.21    1.72 ^ _07431_/Y (sky130_fd_sc_hd__a31oi_1)
   0.18    1.90 v _07553_/Y (sky130_fd_sc_hd__o2111ai_1)
   0.39    2.29 ^ _07920_/Y (sky130_fd_sc_hd__a21boi_0)
   0.07    2.35 v _07923_/Y (sky130_fd_sc_hd__nor3_1)
   0.19    2.54 ^ _07924_/Y (sky130_fd_sc_hd__xnor2_1)
   0.10    2.64 v _07934_/Y (sky130_fd_sc_hd__a221oi_1)
   0.16    2.81 v place248/X (sky130_fd_sc_hd__buf_4)
   0.12    2.92 v _09825_/X (sky130_fd_sc_hd__and2_1)
   0.32    3.24 ^ _10170_/Y (sky130_fd_sc_hd__mux2i_1)
   0.00    3.24 ^ core.CPU_src2_value_a3[20]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           3.24   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.57   11.57   clock network delay (propagated)
   0.00   11.57   clock reconvergence pessimism
          11.57 ^ core.CPU_src2_value_a3[20]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.07   11.50   library setup time
          11.50   data required time
---------------------------------------------------------
          11.50   data required time
          -3.24   data arrival time
---------------------------------------------------------
           8.26   slack (MET)


Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.54    0.54   clock network delay (propagated)
   0.00    0.54 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.19    0.73 ^ core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.73 ^ core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_4)
           0.73   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.54    0.54   clock network delay (propagated)
   0.00    0.54   clock reconvergence pessimism
           0.54 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_4)
  -0.02    0.53   library hold time
           0.53   data required time
---------------------------------------------------------
           0.53   data required time
          -0.73   data arrival time
---------------------------------------------------------
           0.21   slack (MET)


wns max 0.00
tns max 0.00
worst slack max 8.26
```

```
worst slack min 0.21
```

Corner: ff_n40C_1v56

```
Startpoint: core.CPU_src1_value_a3[15]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.88    0.88   clock network delay (propagated)
   0.00    0.88 ^ core.CPU_src1_value_a3[15]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.57    1.45 ^ core.CPU_src1_value_a3[15]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.46    1.91 ^ _10768_/SUM (sky130_fd_sc_hd__ha_1)
   0.11    2.02 v _07734_/Y (sky130_fd_sc_hd__inv_1)
   0.32    2.35 ^ _07763_/Y (sky130_fd_sc_hd__a21oi_1)
   0.15    2.49 v _08107_/Y (sky130_fd_sc_hd__a21oi_1)
   0.58    3.08 ^ _08111_/Y (sky130_fd_sc_hd__o32ai_1)
   0.22    3.30 v _08305_/Y (sky130_fd_sc_hd__a41oi_1)
   0.18    3.48 v _08306_/Y (sky130_fd_sc_hd__xnor2_1)
   0.61    4.09 ^ _08309_/Y (sky130_fd_sc_hd__o221ai_1)
   0.21    4.30 v _08317_/Y (sky130_fd_sc_hd__o22ai_1)
   0.83    5.14 ^ _08319_/Y (sky130_fd_sc_hd__o211ai_1)
   0.22    5.35 v _09183_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00    5.35 v core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           5.35   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.88   11.88   clock network delay (propagated)
   0.00   11.88   clock reconvergence pessimism
          11.88 ^ core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.17   11.71   library setup time
          11.71   data required time
---------------------------------------------------------
          11.71   data required time
          -5.35   data arrival time
---------------------------------------------------------
           6.35   slack (MET)


Startpoint: core.CPU_Xreg_value_a5[17][3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.OUT[3]$_DFF_P_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.83    0.83   clock network delay (propagated)
   0.00    0.83 ^ core.CPU_Xreg_value_a5[17][3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.30    1.13 v core.CPU_Xreg_value_a5[17][3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    1.13 v core.OUT[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           1.13   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.88    0.88   clock network delay (propagated)
   0.00    0.88   clock reconvergence pessimism
           0.88 ^ core.OUT[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.05    0.82   library hold time
           0.82   data required time
---------------------------------------------------------
           0.82   data required time
          -1.13   data arrival time
---------------------------------------------------------
           0.31   slack (MET)


wns max 0.00
tns max 0.00
worst slack max 6.35
worst slack min 0.31
```

Corner: ff_n40C_1v65

```
Startpoint: core.CPU_src1_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[10][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.73    0.73   clock network delay (propagated)
   0.00    0.73 ^ core.CPU_src1_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.36    1.08 ^ core.CPU_src1_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.08    1.16 v _05408_/Y (sky130_fd_sc_hd__inv_1)
   0.22    1.38 v _10738_/COUT (sky130_fd_sc_hd__ha_1)
   0.41    1.79 v _10735_/COUT (sky130_fd_sc_hd__fa_1)
   0.18    1.96 ^ _07426_/Y (sky130_fd_sc_hd__nor2b_1)
   0.09    2.06 v _07428_/Y (sky130_fd_sc_hd__o21bai_1)
   0.33    2.38 ^ _07431_/Y (sky130_fd_sc_hd__a31oi_1)
   0.21    2.59 v _07553_/Y (sky130_fd_sc_hd__o2111ai_1)
   0.56    3.15 ^ _07920_/Y (sky130_fd_sc_hd__a21boi_0)
   0.16    3.31 v _08131_/Y (sky130_fd_sc_hd__o21ai_0)
   0.44    3.75 ^ _08146_/Y (sky130_fd_sc_hd__a211oi_1)
   0.34    4.09 ^ _08147_/X (sky130_fd_sc_hd__a22o_1)
   0.44    4.53 ^ _08149_/X (sky130_fd_sc_hd__o21a_1)
   0.11    4.63 v _08151_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00    4.63 v core.CPU_Xreg_value_a4[10][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           4.63   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.77   11.77   clock network delay (propagated)
   0.00   11.77   clock reconvergence pessimism
          11.77 ^ core.CPU_Xreg_value_a4[10][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.12   11.65   library setup time
          11.65   data required time
---------------------------------------------------------
          11.65   data required time
          -4.63   data arrival time
---------------------------------------------------------
           7.01   slack (MET)


Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.73    0.73   clock network delay (propagated)
   0.00    0.73 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.25    0.98 ^ core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.98 ^ core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_4)
           0.98   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.73    0.73   clock network delay (propagated)
   0.00    0.73   clock reconvergence pessimism
           0.73 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_4)
  -0.02    0.71   library hold time
           0.71   data required time
---------------------------------------------------------
           0.71   data required time
          -0.98   data arrival time
---------------------------------------------------------
           0.27   slack (MET)


wns max 0.00
tns max 0.00
worst slack max 7.01
worst slack min 0.27
```

Corner: ff_n40C_1v76

```
Startpoint: core.CPU_src1_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[10][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.64    0.64   clock network delay (propagated)
   0.00    0.64 ^ core.CPU_src1_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.31    0.95 ^ core.CPU_src1_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.07    1.02 v _05408_/Y (sky130_fd_sc_hd__inv_1)
   0.19    1.22 v _10738_/COUT (sky130_fd_sc_hd__ha_1)
   0.35    1.57 v _10735_/COUT (sky130_fd_sc_hd__fa_1)
   0.15    1.72 ^ _07426_/Y (sky130_fd_sc_hd__nor2b_1)
   0.08    1.81 v _07428_/Y (sky130_fd_sc_hd__o21bai_1)
   0.28    2.09 ^ _07431_/Y (sky130_fd_sc_hd__a31oi_1)
   0.18    2.27 v _07553_/Y (sky130_fd_sc_hd__o2111ai_1)
   0.49    2.76 ^ _07920_/Y (sky130_fd_sc_hd__a21boi_0)
   0.12    2.88 v _08131_/Y (sky130_fd_sc_hd__o21ai_0)
   0.37    3.25 ^ _08146_/Y (sky130_fd_sc_hd__a211oi_1)
   0.29    3.54 ^ _08147_/X (sky130_fd_sc_hd__a22o_1)
   0.38    3.92 ^ _08149_/X (sky130_fd_sc_hd__o21a_1)
   0.08    4.00 v _08151_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00    4.00 v core.CPU_Xreg_value_a4[10][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           4.00   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.68   11.68   clock network delay (propagated)
   0.00   11.68   clock reconvergence pessimism
          11.68 ^ core.CPU_Xreg_value_a4[10][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.10   11.57   library setup time
          11.57   data required time
---------------------------------------------------------
          11.57   data required time
          -4.00   data arrival time
---------------------------------------------------------
           7.57   slack (MET)


Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.64    0.64   clock network delay (propagated)
   0.00    0.64 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.22    0.87 ^ core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.87 ^ core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_4)
           0.87   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.64    0.64   clock network delay (propagated)
   0.00    0.64   clock reconvergence pessimism
           0.64 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_4)
  -0.02    0.63   library hold time
           0.63   data required time
---------------------------------------------------------
           0.63   data required time
          -0.87   data arrival time
---------------------------------------------------------
           0.24   slack (MET)


wns max 0.00
tns max 0.00
worst slack max 7.57
worst slack min 0.24
```

Corner: ff_n40C_1v95

```
Startpoint: core.CPU_src1_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[10][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.54    0.54   clock network delay (propagated)
   0.00    0.54 ^ core.CPU_src1_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.26    0.81 ^ core.CPU_src1_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.06    0.86 v _05408_/Y (sky130_fd_sc_hd__inv_1)
   0.17    1.03 v _10738_/COUT (sky130_fd_sc_hd__ha_1)
   0.29    1.32 v _10735_/COUT (sky130_fd_sc_hd__fa_1)
   0.13    1.44 ^ _07426_/Y (sky130_fd_sc_hd__nor2b_1)
   0.07    1.51 v _07428_/Y (sky130_fd_sc_hd__o21bai_1)
   0.23    1.75 ^ _07431_/Y (sky130_fd_sc_hd__a31oi_1)
   0.15    1.89 v _07553_/Y (sky130_fd_sc_hd__o2111ai_1)
   0.40    2.30 ^ _07920_/Y (sky130_fd_sc_hd__a21boi_0)
   0.09    2.38 v _08131_/Y (sky130_fd_sc_hd__o21ai_0)
   0.30    2.68 ^ _08146_/Y (sky130_fd_sc_hd__a211oi_1)
   0.23    2.91 ^ _08147_/X (sky130_fd_sc_hd__a22o_1)
   0.31    3.22 ^ _08149_/X (sky130_fd_sc_hd__o21a_1)
   0.06    3.28 v _08151_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00    3.28 v core.CPU_Xreg_value_a4[10][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           3.28   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.57   11.57   clock network delay (propagated)
   0.00   11.57   clock reconvergence pessimism
          11.57 ^ core.CPU_Xreg_value_a4[10][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.08   11.49   library setup time
          11.49   data required time
---------------------------------------------------------
          11.49   data required time
          -3.28   data arrival time
---------------------------------------------------------
           8.21   slack (MET)


Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.54    0.54   clock network delay (propagated)
   0.00    0.54 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.18    0.73 ^ core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.73 ^ core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_4)
           0.73   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.54    0.54   clock network delay (propagated)
   0.00    0.54   clock reconvergence pessimism
           0.54 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_4)
  -0.02    0.53   library hold time
           0.53   data required time
---------------------------------------------------------
           0.53   data required time
          -0.73   data arrival time
---------------------------------------------------------
           0.20   slack (MET)


wns max 0.00
tns max 0.00
worst slack max 8.21
worst slack min 0.20
```

Corner: ss_100C_1v40

```
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   1.84    1.84   clock network delay (propagated)
   0.00    1.84 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.94    2.77 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.79    3.57 ^ place449/X (sky130_fd_sc_hd__buf_4)
   0.88    4.45 ^ _10805_/COUT (sky130_fd_sc_hd__ha_1)
   0.43    4.88 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
   1.77    6.65 v _07525_/X (sky130_fd_sc_hd__or3b_2)
   1.85    8.50 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
   1.08    9.58 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
   0.60   10.18 v place292/X (sky130_fd_sc_hd__buf_4)
   0.53   10.71 v _08170_/X (sky130_fd_sc_hd__o21a_1)
   0.61   11.32 ^ _08173_/Y (sky130_fd_sc_hd__o31ai_1)
   0.56   11.88 ^ place264/X (sky130_fd_sc_hd__buf_4)
   0.21   12.10 v _08187_/Y (sky130_fd_sc_hd__a211oi_1)
   1.24   13.34 ^ _08188_/Y (sky130_fd_sc_hd__a221oi_1)
   0.91   14.24 v _08192_/Y (sky130_fd_sc_hd__o221ai_1)
   0.60   14.84 v place246/X (sky130_fd_sc_hd__buf_4)
   0.47   15.31 ^ _09262_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00   15.31 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          15.31   data arrival time

  11.00   11.00   clock clk (rise edge)
   1.94   12.94   clock network delay (propagated)
   0.00   12.94   clock reconvergence pessimism
          12.94 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.28   12.66   library setup time
          12.66   data required time
---------------------------------------------------------
          12.66   data required time
         -15.31   data arrival time
---------------------------------------------------------
          -2.64   slack (VIOLATED)


Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   1.85    1.85   clock network delay (propagated)
   0.00    1.85 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.81    2.66 v core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    2.66 v core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_4)
           2.66   data arrival time

   0.00    0.00   clock clk (rise edge)
   1.85    1.85   clock network delay (propagated)
   0.00    1.85   clock reconvergence pessimism
           1.85 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_4)
  -0.10    1.75   library hold time
           1.75   data required time
---------------------------------------------------------
           1.75   data required time
          -2.66   data arrival time
---------------------------------------------------------
           0.91   slack (MET)


wns max -2.64
tns max -340.90
```

```
worst slack max -2.64
worst slack min 0.91
```

Corner: ss_100C_1v60

```
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   1.33    1.33   clock network delay (propagated)
   0.00    1.33 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.58    1.92 v core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.44    2.36 v place449/X (sky130_fd_sc_hd__buf_4)
   0.79    3.15 ^ _10805_/SUM (sky130_fd_sc_hd__ha_1)
   0.32    3.47 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
   1.13    4.60 v _07525_/X (sky130_fd_sc_hd__or3b_2)
   1.27    5.86 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
   0.66    6.52 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
   0.42    6.95 v place292/X (sky130_fd_sc_hd__buf_4)
   0.37    7.31 v _08170_/X (sky130_fd_sc_hd__o21a_1)
   0.42    7.73 ^ _08173_/Y (sky130_fd_sc_hd__o31ai_1)
   0.38    8.11 ^ place264/X (sky130_fd_sc_hd__buf_4)
   0.15    8.26 v _08187_/Y (sky130_fd_sc_hd__a211oi_1)
   0.86    9.13 ^ _08188_/Y (sky130_fd_sc_hd__a221oi_1)
   0.59    9.71 v _08192_/Y (sky130_fd_sc_hd__o221ai_1)
   0.43   10.14 v place246/X (sky130_fd_sc_hd__buf_4)
   0.33   10.47 ^ _09262_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00   10.47 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          10.47   data arrival time

  11.00   11.00   clock clk (rise edge)
   1.41   12.41   clock network delay (propagated)
   0.00   12.41   clock reconvergence pessimism
          12.41 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.18   12.23   library setup time
          12.23   data required time
---------------------------------------------------------
          12.23   data required time
         -10.47   data arrival time
---------------------------------------------------------
           1.76   slack (MET)


Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   1.34    1.34   clock network delay (propagated)
   0.00    1.34 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.57    1.91 v core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    1.91 v core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_4)
           1.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   1.34    1.34   clock network delay (propagated)
   0.00    1.34   clock reconvergence pessimism
           1.34 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_4)
  -0.08    1.26   library hold time
           1.26   data required time
---------------------------------------------------------
           1.26   data required time
          -1.91   data arrival time
---------------------------------------------------------
           0.65   slack (MET)


wns max 0.00
tns max 0.00
```

```
worst slack max 1.76
worst slack min 0.65
```

Corner: ss_n40C_1v28

```
Startpoint: core.CPU_src1_value_a3[13]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   5.17    5.17   clock network delay (propagated)
   0.00    5.17 ^ core.CPU_src1_value_a3[13]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   3.61    8.78 ^ core.CPU_src1_value_a3[13]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   2.66   11.44 ^ _10764_/SUM (sky130_fd_sc_hd__ha_1)
   0.91   12.35 v _08104_/Y (sky130_fd_sc_hd__nor2b_1)
   2.24   14.60 ^ _08105_/Y (sky130_fd_sc_hd__o21ai_0)
   1.58   16.18 v _08107_/Y (sky130_fd_sc_hd__a21oi_1)
   4.61   20.79 ^ _08111_/Y (sky130_fd_sc_hd__o32ai_1)
   2.59   23.38 v _08305_/Y (sky130_fd_sc_hd__a41oi_1)
   1.14   24.52 v _08306_/Y (sky130_fd_sc_hd__xnor2_1)
   4.78   29.29 ^ _08309_/Y (sky130_fd_sc_hd__o221ai_1)
   3.03   32.33 v _08317_/Y (sky130_fd_sc_hd__o22ai_1)
   5.77   38.09 ^ _08319_/Y (sky130_fd_sc_hd__o211ai_1)
   3.28   41.38 v _09183_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00   41.38 v core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          41.38   data arrival time

  11.00   11.00   clock clk (rise edge)
   5.11   16.11   clock network delay (propagated)
   0.00   16.11   clock reconvergence pessimism
          16.11 ^ core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -1.76   14.35   library setup time
          14.35   data required time
---------------------------------------------------------
          14.35   data required time
         -41.38   data arrival time
---------------------------------------------------------
         -27.02   slack (VIOLATED)


Startpoint: core.CPU_Xreg_value_a5[17][3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.OUT[3]$_DFF_P_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   4.80    4.80   clock network delay (propagated)
   0.00    4.80 ^ core.CPU_Xreg_value_a5[17][3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.71    6.51 v core.CPU_Xreg_value_a5[17][3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    6.51 v core.OUT[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           6.51   data arrival time

   0.00    0.00   clock clk (rise edge)
   5.12    5.12   clock network delay (propagated)
   0.00    5.12   clock reconvergence pessimism
           5.12 ^ core.OUT[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.28    4.85   library hold time
           4.85   data required time
---------------------------------------------------------
           4.85   data required time
          -6.51   data arrival time
---------------------------------------------------------
           1.67   slack (MET)


wns max -27.02
tns max -12774.20
worst slack max -27.02
worst slack min 1.67
```

Corner: ss_n40C_1v35

```
Startpoint: core.CPU_src1_value_a3[13]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   3.66    3.66   clock network delay (propagated)
   0.00    3.66 ^ core.CPU_src1_value_a3[13]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   2.56    6.21 ^ core.CPU_src1_value_a3[13]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.95    8.16 ^ _10764_/SUM (sky130_fd_sc_hd__ha_1)
   0.64    8.80 v _08104_/Y (sky130_fd_sc_hd__nor2b_1)
   1.57   10.37 ^ _08105_/Y (sky130_fd_sc_hd__o21ai_0)
   1.07   11.44 v _08107_/Y (sky130_fd_sc_hd__a21oi_1)
   3.10   14.54 ^ _08111_/Y (sky130_fd_sc_hd__o32ai_1)
   1.73   16.27 v _08305_/Y (sky130_fd_sc_hd__a41oi_1)
   0.77   17.04 v _08306_/Y (sky130_fd_sc_hd__xnor2_1)
   3.15   20.19 ^ _08309_/Y (sky130_fd_sc_hd__o221ai_1)
   1.93   22.12 v _08317_/Y (sky130_fd_sc_hd__o22ai_1)
   3.97   26.09 ^ _08319_/Y (sky130_fd_sc_hd__o211ai_1)
   2.05   28.14 v _09183_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00   28.14 v core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          28.14   data arrival time

  11.00   11.00   clock clk (rise edge)
   3.62   14.62   clock network delay (propagated)
   0.00   14.62   clock reconvergence pessimism
          14.62 ^ core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -1.17   13.44   library setup time
          13.44   data required time
---------------------------------------------------------
          13.44   data required time
         -28.14   data arrival time
---------------------------------------------------------
         -14.70   slack (VIOLATED)


Startpoint: core.CPU_Xreg_value_a5[17][3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.OUT[3]$_DFF_P_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   3.40    3.40   clock network delay (propagated)
   0.00    3.40 ^ core.CPU_Xreg_value_a5[17][3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.27    4.67 v core.CPU_Xreg_value_a5[17][3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    4.67 v core.OUT[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           4.67   data arrival time

   0.00    0.00   clock clk (rise edge)
   3.62    3.62   clock network delay (propagated)
   0.00    3.62   clock reconvergence pessimism
           3.62 ^ core.OUT[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.21    3.41   library hold time
           3.41   data required time
---------------------------------------------------------
           3.41   data required time
          -4.67   data arrival time
---------------------------------------------------------
           1.26   slack (MET)


wns max -14.70
tns max -5623.47
worst slack max -14.70
worst slack min 1.26
```

Corner: ss_n40C_1v40

```
Startpoint: core.CPU_src1_value_a3[13]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   2.99    2.99   clock network delay (propagated)
   0.00    2.99 ^ core.CPU_src1_value_a3[13]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   2.09    5.08 ^ core.CPU_src1_value_a3[13]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.63    6.71 ^ _10764_/SUM (sky130_fd_sc_hd__ha_1)
   0.52    7.23 v _08104_/Y (sky130_fd_sc_hd__nor2b_1)
   1.27    8.50 ^ _08105_/Y (sky130_fd_sc_hd__o21ai_0)
   0.86    9.36 v _08107_/Y (sky130_fd_sc_hd__a21oi_1)
   2.46   11.82 ^ _08111_/Y (sky130_fd_sc_hd__o32ai_1)
   1.34   13.16 v _08305_/Y (sky130_fd_sc_hd__a41oi_1)
   0.64   13.80 v _08306_/Y (sky130_fd_sc_hd__xnor2_1)
   2.48   16.28 ^ _08309_/Y (sky130_fd_sc_hd__o221ai_1)
   1.50   17.78 v _08317_/Y (sky130_fd_sc_hd__o22ai_1)
   3.19   20.97 ^ _08319_/Y (sky130_fd_sc_hd__o211ai_1)
   1.60   22.56 v _09183_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00   22.56 v core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          22.56   data arrival time

  11.00   11.00   clock clk (rise edge)
   2.96   13.96   clock network delay (propagated)
   0.00   13.96   clock reconvergence pessimism
          13.96 ^ core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.92   13.04   library setup time
          13.04   data required time
---------------------------------------------------------
          13.04   data required time
         -22.56   data arrival time
---------------------------------------------------------
          -9.52   slack (VIOLATED)


Startpoint: core.CPU_Xreg_value_a5[17][3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.OUT[3]$_DFF_P_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   2.79    2.79   clock network delay (propagated)
   0.00    2.79 ^ core.CPU_Xreg_value_a5[17][3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.06    3.85 v core.CPU_Xreg_value_a5[17][3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    3.85 v core.OUT[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           3.85   data arrival time

   0.00    0.00   clock clk (rise edge)
   2.97    2.97   clock network delay (propagated)
   0.00    2.97   clock reconvergence pessimism
           2.97 ^ core.OUT[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.18    2.79   library hold time
           2.79   data required time
---------------------------------------------------------
           2.79   data required time
          -3.85   data arrival time
---------------------------------------------------------
           1.06   slack (MET)


wns max -9.52
tns max -2787.18
worst slack max -9.52
worst slack min 1.06
```

Corner: ss_n40C_1v44

```
Startpoint: core.CPU_src1_value_a3[13]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   2.59    2.59   clock network delay (propagated)
   0.00    2.59 ^ core.CPU_src1_value_a3[13]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.81    4.41 ^ core.CPU_src1_value_a3[13]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.42    5.82 ^ _10764_/SUM (sky130_fd_sc_hd__ha_1)
   0.44    6.27 v _08104_/Y (sky130_fd_sc_hd__nor2b_1)
   1.10    7.36 ^ _08105_/Y (sky130_fd_sc_hd__o21ai_0)
   0.73    8.09 v _08107_/Y (sky130_fd_sc_hd__a21oi_1)
   2.09   10.18 ^ _08111_/Y (sky130_fd_sc_hd__o32ai_1)
   1.14   11.32 v _08305_/Y (sky130_fd_sc_hd__a41oi_1)
   0.59   11.91 v _08306_/Y (sky130_fd_sc_hd__xnor2_1)
   2.10   14.01 ^ _08309_/Y (sky130_fd_sc_hd__o221ai_1)
   1.25   15.26 v _08317_/Y (sky130_fd_sc_hd__o22ai_1)
   2.74   18.00 ^ _08319_/Y (sky130_fd_sc_hd__o211ai_1)
   1.31   19.31 v _09183_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00   19.31 v core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          19.31   data arrival time

  11.00   11.00   clock clk (rise edge)
   2.57   13.57   clock network delay (propagated)
   0.00   13.57   clock reconvergence pessimism
          13.57 ^ core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.76   12.81   library setup time
          12.81   data required time
---------------------------------------------------------
          12.81   data required time
         -19.31   data arrival time
---------------------------------------------------------
          -6.50   slack (VIOLATED)


Startpoint: core.CPU_Xreg_value_a5[17][3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.OUT[3]$_DFF_P_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   2.43    2.43   clock network delay (propagated)
   0.00    2.43 ^ core.CPU_Xreg_value_a5[17][3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.93    3.36 v core.CPU_Xreg_value_a5[17][3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    3.36 v core.OUT[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           3.36   data arrival time

   0.00    0.00   clock clk (rise edge)
   2.57    2.57   clock network delay (propagated)
   0.00    2.57   clock reconvergence pessimism
           2.57 ^ core.OUT[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.16    2.41   library hold time
           2.41   data required time
---------------------------------------------------------
           2.41   data required time
          -3.36   data arrival time
---------------------------------------------------------
           0.95   slack (MET)


wns max -6.50
tns max -1500.01
worst slack max -6.50
worst slack min 0.95
```

Corner: ss_n40C_1v60

```
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   1.59    1.59   clock network delay (propagated)
   0.00    1.59 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.69    2.29 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.71    3.00 ^ place449/X (sky130_fd_sc_hd__buf_4)
   0.72    3.72 ^ _10805_/COUT (sky130_fd_sc_hd__ha_1)
   0.29    4.01 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
   1.29    5.30 v _07525_/X (sky130_fd_sc_hd__or3b_2)
   1.51    6.81 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
   0.76    7.57 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
   0.46    8.03 v place292/X (sky130_fd_sc_hd__buf_4)
   0.43    8.46 v _08170_/X (sky130_fd_sc_hd__o21a_1)
   0.58    9.04 ^ _08173_/Y (sky130_fd_sc_hd__o31ai_1)
   0.46    9.50 ^ place264/X (sky130_fd_sc_hd__buf_4)
   0.12    9.63 v _08187_/Y (sky130_fd_sc_hd__a211oi_1)
   1.20   10.83 ^ _08188_/Y (sky130_fd_sc_hd__a221oi_1)
   0.65   11.47 v _08192_/Y (sky130_fd_sc_hd__o221ai_1)
   0.46   11.93 v place246/X (sky130_fd_sc_hd__buf_4)
   0.37   12.31 ^ _09262_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00   12.31 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          12.31   data arrival time

  11.00   11.00   clock clk (rise edge)
   1.70   12.70   clock network delay (propagated)
   0.00   12.70   clock reconvergence pessimism
          12.70 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.19   12.51   library setup time
          12.51   data required time
---------------------------------------------------------
          12.51   data required time
         -12.31   data arrival time
---------------------------------------------------------
           0.20   slack (MET)


Startpoint: core.CPU_Xreg_value_a5[17][3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.OUT[3]$_DFF_P_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   1.60    1.60   clock network delay (propagated)
   0.00    1.60 ^ core.CPU_Xreg_value_a5[17][3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.63    2.24 v core.CPU_Xreg_value_a5[17][3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    2.24 v core.OUT[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           2.24   data arrival time

   0.00    0.00   clock clk (rise edge)
   1.70    1.70   clock network delay (propagated)
   0.00    1.70   clock reconvergence pessimism
           1.70 ^ core.OUT[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.11    1.59   library hold time
           1.59   data required time
---------------------------------------------------------
           1.59   data required time
          -2.24   data arrival time
---------------------------------------------------------
           0.64   slack (MET)


wns max 0.00
tns max 0.00
worst slack max 0.20
```

worst slack min 0.64

Corner: ss_n40C_1v76

```
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   1.20    1.20   clock network delay (propagated)
   0.00    1.20 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.50    1.70 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.53    2.23 ^ place449/X (sky130_fd_sc_hd__buf_4)
   0.52    2.75 ^ _10805_/COUT (sky130_fd_sc_hd__ha_1)
   0.22    2.97 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
   0.88    3.85 v _07525_/X (sky130_fd_sc_hd__or3b_2)
   1.08    4.93 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
   0.49    5.42 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
   0.35    5.77 v place292/X (sky130_fd_sc_hd__buf_4)
   0.32    6.08 v _08170_/X (sky130_fd_sc_hd__o21a_1)
   0.41    6.49 ^ _08173_/Y (sky130_fd_sc_hd__o31ai_1)
   0.32    6.81 ^ place264/X (sky130_fd_sc_hd__buf_4)
   0.10    6.91 v _08187_/Y (sky130_fd_sc_hd__a211oi_1)
   0.87    7.77 ^ _08188_/Y (sky130_fd_sc_hd__a221oi_1)
   0.44    8.22 v _08192_/Y (sky130_fd_sc_hd__o221ai_1)
   0.36    8.57 v place246/X (sky130_fd_sc_hd__buf_4)
   0.27    8.85 ^ _09262_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00    8.85 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           8.85   data arrival time

  11.00   11.00   clock clk (rise edge)
   1.27   12.27   clock network delay (propagated)
   0.00   12.27   clock reconvergence pessimism
          12.27 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.14   12.13   library setup time
          12.13   data required time
---------------------------------------------------------
          12.13   data required time
          -8.85   data arrival time
---------------------------------------------------------
           3.28   slack (MET)


Startpoint: core.CPU_Xreg_value_a5[17][3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.OUT[3]$_DFF_P_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   1.21    1.21   clock network delay (propagated)
   0.00    1.21 ^ core.CPU_Xreg_value_a5[17][3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.48    1.68 v core.CPU_Xreg_value_a5[17][3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    1.68 v core.OUT[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           1.68   data arrival time

   0.00    0.00   clock clk (rise edge)
   1.27    1.27   clock network delay (propagated)
   0.00    1.27   clock reconvergence pessimism
           1.27 ^ core.OUT[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.08    1.19   library hold time
           1.19   data required time
---------------------------------------------------------
           1.19   data required time
          -1.68   data arrival time
---------------------------------------------------------
           0.49   slack (MET)


wns max 0.00
tns max 0.00
worst slack max 3.28
```

```
worst slack min 0.49
```

Corner: tt_025C_1v80

```
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.78    0.78   clock network delay (propagated)
   0.00    0.78 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.31    1.09 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.34    1.43 ^ place449/X (sky130_fd_sc_hd__buf_4)
   0.31    1.74 ^ _10805_/COUT (sky130_fd_sc_hd__ha_1)
   0.15    1.88 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
   0.48    2.36 v _07525_/X (sky130_fd_sc_hd__or3b_2)
   0.67    3.03 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
   0.24    3.27 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
   0.22    3.49 v place292/X (sky130_fd_sc_hd__buf_4)
   0.19    3.68 v _08170_/X (sky130_fd_sc_hd__o21a_1)
   0.23    3.92 ^ _08173_/Y (sky130_fd_sc_hd__o31ai_1)
   0.18    4.10 ^ place264/X (sky130_fd_sc_hd__buf_4)
   0.07    4.17 v _08187_/Y (sky130_fd_sc_hd__a211oi_1)
   0.51    4.68 ^ _08188_/Y (sky130_fd_sc_hd__a221oi_1)
   0.24    4.92 v _08192_/Y (sky130_fd_sc_hd__o221ai_1)
   0.23    5.15 v place246/X (sky130_fd_sc_hd__buf_4)
   0.18    5.33 ^ _09262_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00    5.33 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           5.33   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.83   11.83   clock network delay (propagated)
   0.00   11.83   clock reconvergence pessimism
          11.83 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.08   11.75   library setup time
          11.75   data required time
---------------------------------------------------------
          11.75   data required time
          -5.33   data arrival time
---------------------------------------------------------
           6.42   slack (MET)


Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.79    0.79   clock network delay (propagated)
   0.00    0.79 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.29    1.08 v core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    1.08 v core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_4)
           1.08   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.79    0.79   clock network delay (propagated)
   0.00    0.79   clock reconvergence pessimism
           0.79 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_4)
  -0.03    0.75   library hold time
           0.75   data required time
---------------------------------------------------------
           0.75   data required time
          -1.08   data arrival time
---------------------------------------------------------
           0.32   slack (MET)


wns max 0.00
tns max 0.00
```

```
worst slack max 6.42
worst slack min 0.32
```

Corner: tt_100C_1v80

```
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.76    0.76   clock network delay (propagated)
   0.00    0.76 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.30    1.07 v core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.25    1.31 v place449/X (sky130_fd_sc_hd__buf_4)
   0.40    1.72 ^ _10805_/SUM (sky130_fd_sc_hd__ha_1)
   0.16    1.88 v _07452_/Y (sky130_fd_sc_hd__o21ai_0)
   0.47    2.35 v _07525_/X (sky130_fd_sc_hd__or3b_2)
   0.65    2.99 ^ _07531_/Y (sky130_fd_sc_hd__o21ai_0)
   0.24    3.24 v _07955_/Y (sky130_fd_sc_hd__a221oi_1)
   0.22    3.46 v place292/X (sky130_fd_sc_hd__buf_4)
   0.19    3.65 v _08170_/X (sky130_fd_sc_hd__o21a_1)
   0.21    3.86 ^ _08173_/Y (sky130_fd_sc_hd__o31ai_1)
   0.18    4.04 ^ place264/X (sky130_fd_sc_hd__buf_4)
   0.08    4.12 v _08187_/Y (sky130_fd_sc_hd__a211oi_1)
   0.47    4.58 ^ _08188_/Y (sky130_fd_sc_hd__a221oi_1)
   0.25    4.83 v _08192_/Y (sky130_fd_sc_hd__o221ai_1)
   0.23    5.07 v place246/X (sky130_fd_sc_hd__buf_4)
   0.17    5.24 ^ _09262_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00    5.24 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           5.24   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.81   11.81   clock network delay (propagated)
   0.00   11.81   clock reconvergence pessimism
          11.81 ^ core.CPU_Xreg_value_a4[2][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.08   11.73   library setup time
          11.73   data required time
---------------------------------------------------------
          11.73   data required time
          -5.24   data arrival time
---------------------------------------------------------
           6.48   slack (MET)


Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.77    0.77   clock network delay (propagated)
   0.00    0.77 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.30    1.06 v core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    1.06 v core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_4)
           1.06   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.77    0.77   clock network delay (propagated)
   0.00    0.77   clock reconvergence pessimism
           0.77 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_4)
  -0.03    0.74   library hold time
           0.74   data required time
---------------------------------------------------------
           0.74   data required time
          -1.06   data arrival time
---------------------------------------------------------
           0.32   slack (MET)


wns max 0.00
tns max 0.00
```

```
worst slack max 6.48
worst slack min 0.32
```